

Acids, bases and salts

Igcse Chemistry

Acids

An **acid** 酸 is a substance that forms hydrogen ions (H^+) when dissolved in water. Acids have three typical reactions:

- with **metals** 金属: acid + metal $\rightarrow$ a **salt** 盐 + **hydrogen** 氢气
- with bases: acid + **base** 碱 $\rightarrow$ a salt + water (this is **neutralisation** 中和)
- with **carbonates** 碳酸盐: acid + carbonate $\rightarrow$ a salt + water + **carbon dioxide** 二氧化碳

Indicators

An **indicator** 指示剂 is a dye that changes colour to show whether a solution is acidic or alkaline.

Indicator	In acid	In alkali
litmus 石蕊	red	blue
thymolphthalein 百里酚酞	colourless	blue
methyl orange 甲基橙	red	yellow

Aqueous solutions of acids contain hydrogen **ions** 离子 (H^+). Aqueous solutions of alkalis contain hydroxide ions (OH^-).

Bases and alkalis

Bases are **oxides** 氧化物 or **hydroxides** 氢氧化物 of metals. An **alkali** 可溶性碱 is a base that dissolves in water (a soluble base).

Bases have two typical reactions:

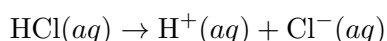
- with acids: base + acid $\rightarrow$ a salt + water (neutralisation again)
- with **ammonium salts** 铵盐: this releases ammonia gas.

Strong and weak acids

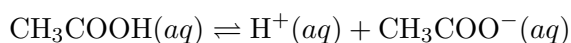
An acid is a **proton** 质子 donor —it gives away H^+ ions (a hydrogen ion is just a proton). A base is a proton acceptor.

How strong an acid is depends on how much of it splits into ions in water:

- A **strong acid** 强酸 is completely **dissociated** 电离 (fully split into ions) in water. **Hydrochloric acid** 盐酸 is strong:



- A **weak acid** 弱酸 is only partly dissociated. **Ethanoic acid** 乙酸 is weak, so the equation uses the reversible arrow:



Note: 'strong' and 'weak' are about dissociation, not about being dilute or concentrated.

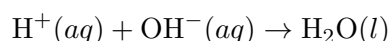
The pH scale

The pH scale runs from 0 to 14 and tells you how acidic or alkaline a solution is. A higher hydrogen ion concentration means a lower pH.

You can find pH using **universal indicator** 通用指示剂, which turns different colours:

pH	Type	Colour of universal indicator
below 7	acidic 酸性	red / orange / yellow
exactly 7	neutral 中性	green
above 7	alkaline 碱性	blue / purple

When an acid and an alkali react, the H^+ and OH^- ions join to make water:



Oxides

Oxides can be sorted by how they behave:

- **Acidic oxides** are oxides of non-metals, such as SO_2 and CO_2 .
- **Basic oxides** are oxides of metals, such as CuO and CaO .
- **Amphoteric** 两性 oxides react with both acids and bases to make a salt and water. Al_2O_3 and ZnO are amphoteric.

So metal oxides tend to be basic and non-metal oxides tend to be acidic.

Preparing salts

Whether a salt can be made by a certain method depends on whether it is **soluble** 可溶 (dissolves) or **insoluble** 不溶 (does not dissolve).

Solubility rules

Salt type	Rule
sodium, potassium, ammonium salts	all soluble
nitrates 硝酸盐	all soluble
chlorides 氯化物	soluble, except lead and silver
sulfates 硫酸盐	soluble, except barium, calcium and lead
carbonates	insoluble, except sodium, potassium and ammonium
hydroxides	insoluble, except sodium, potassium, ammonium and (partly) calcium

Making a soluble salt

You react an acid with one of these:

- an **alkali**, using **titration** 滴定 (since both are solutions, you must measure the exact volumes);
- an **excess** 过量 of a metal, an insoluble base, or an insoluble carbonate.

When you use an excess of a solid, you then **filter** 过滤 to remove the leftover solid. Finally you **evaporate** 蒸发 some water and let the solution **crystallise** 结晶 to get the salt.

Making an insoluble salt

An insoluble salt is made by **precipitation** 沉淀: mix two solutions that each contain one of the needed ions, and the insoluble salt forms as a solid. You then filter, wash and dry it.

Water in salts

- A **hydrated** 水合 substance is chemically joined with water.
- An **anhydrous** 无水 substance contains no water.

The water molecules inside hydrated crystals are called the **water of crystallisation** 结晶水. For example, $\text{CuSO}_4 \bullet 5\text{H}_2\text{O}$ has five water molecules for each formula unit.